

Step 9 - Build the ADF Pipeline

Phase C · Linked Service · Datasets · Copy · Debug · ~20 min

Goal: Build and debug the real pipeline: linked service, source/sink datasets, a date-stamped sink path via ADF expressions, Copy activity with fault tolerance - ending in green checkmarks and 10,000/10,000 rows. Artifacts: pipeline JSON + screenshots.

LEARN

The dependency chain: Linked Service (how to reach the storage account) - Datasets (where exactly the data sits) - Pipeline with Copy activity (the work). One linked service serves both datasets since source and bronze are containers in the same account.

Auth: Account key today (zero-friction lab); the interview answer is Managed Identity - no secrets, Azure AD-audited, OSFI least-privilege. The pro path at the end switches it properly.

Dynamic content: the sink folder uses ADF's expression language to stamp today's date - same `business_date` partitioning convention as your lakehouse.

LAB

9.1 — Linked Service: Manage - Linked services - New - Azure Data Lake Storage Gen2. Name `LS_MapleBank_ADLS`, AutoResolve IR, auth Account key, pick your subscription + storage account. TEST CONNECTION must pass. Create.

9.2 — Source dataset: Author - Datasets - New - ADLS Gen2 - DelimitedText. Name `DS_Source_Transactions_CSV`, linked service `LS_MapleBank_ADLS`, browse to source / `fact_transactions.csv`, First row as header ON, import schema from connection.

9.3 — Sink dataset: same type. Name `DS_Bronze_Transactions`, container `bronze`, directory and file empty, header ON, schema None. Then on its Connection tab, click into Directory - Add dynamic content - paste:

```
@concat('transactions/business_date=', formatDateTime(utcnow(),'yyyy-MM-dd'))
```

File name box: `transactions.csv`

9.4 — Pipeline `PL_MapleBank_Ingest_Transactions`: drag Move & transform - Copy data onto the canvas, name `Copy_Txn_to_Bronze`. Source tab = source dataset; Sink tab = sink dataset; Settings tab - Fault tolerance - Skip incompatible rows. Validate: no errors.

9.5 — Debug: click Debug, watch Output: Queued - In progress - Succeeded. Eyeglasses icon: rows read/written should be 10,000 / 10,000.

9.6 — Verify landing: Portal - storage account - Storage browser - bronze - `transactions/business_date=YYYY-MM-DD/transactions.csv`.

9.7 — Capture the JSON ({} icon, top-right):

```
{
  "name": "PL_MapleBank_Ingest_Transactions",
  "properties": {
    "activities": [{
      "name": "Copy_Txn_to_Bronze",
      "type": "Copy",
      "inputs": [{ "referenceName": "DS_Source_Transactions_CSV",
        "type": "DatasetReference" }],
      "outputs": [{ "referenceName": "DS_Bronze_Transactions",
        "type": "DatasetReference" }],
      "typeProperties": {
        "source": { "type": "DelimitedTextSource" },
        "sink": { "type": "DelimitedTextSink" },
        "enableSkipIncompatibleRow": true
      }
    }
  ]
}
```

Optional Pro Path — Managed Identity (10 min)

Storage account - Access Control (IAM) - Add role assignment - role Storage Blob Data Contributor - assign to Managed identity - select your ADF - save, wait ~2 min. Edit LS_MapleBank_ADLS - auth System Assigned Managed Identity - Test - re-Debug. Now your lab runs production-grade auth.

COMMIT + CHECKPOINT

To adf/: pipeline JSON; screenshots of the green debug run, run details (rows read/written), and the date-stamped bronze folder. Pass criteria: Succeeded run, 10,000/10,000 rows, folder visible. Common failures: Test connection (key/firewall), dynamic content typed directly into the box instead of via Add dynamic content, debug Forbidden (storage firewall).